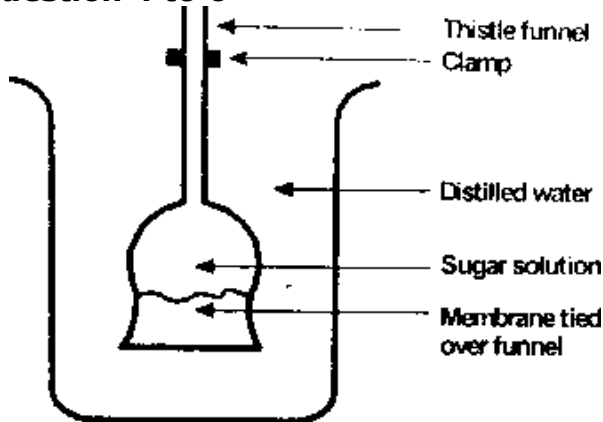


# UTME 2018 BIOLOGY QUESTIONS

1. A group of closely related organisms capable of interbreeding to produce fertile offspring are known as members of a
  - A. kingdom
  - B. class
  - C. family
  - D. species
2. A beaker of pond water containing few specimens of Euglena was placed in a dark room for two weeks. At the end of this period, the specimens of Euglena were still alive because they were
  - A. able to carry out holozoic nutrition
  - B. able to carry out photosynthesis using carbon dioxide in the pond water
  - C. better adapted to life in darkness than to life in light
  - D. not overcrowded
3. The cytoplasm of the cell is considered a very important component because it
  - A. regulates the amount of energy in the cell
  - B. suspends all cell organelles
  - C. is the outermost part of the cell
  - D. is solely responsible for cell division

**Use the diagram below to answer question 4 to 6**



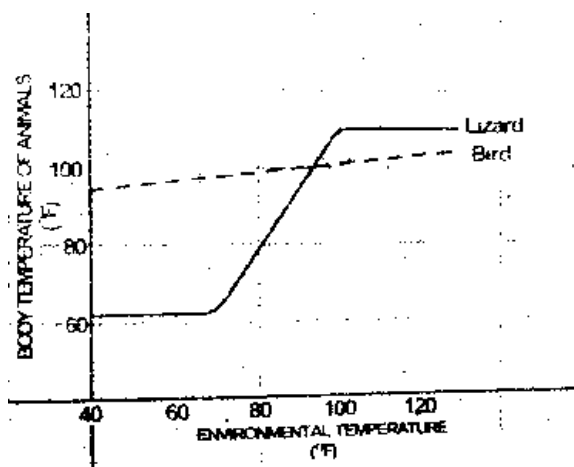
4. After an hour, the level of water in the thistle funnel will
  - A. rise
  - B. fall
  - C. remain the same
  - D. double
5. The experiment is used to demonstrate the process of
  - A. transportation
  - B. water culture
  - C. diffusion
  - D. Osmosis

6. In a plant cells, the role of the membrane is played by the
  - A. nucleolus
  - B. cell wall
  - C. cytoplasm
  - D. mitochondrion
7. Red blood cells were found to have burst open after being placed in distil for an hour. This phenomenon is known as
  - A. plasmolysis
  - B. diffusion
  - C. haemolysis
  - D. wilting
8. The curvature movement of plants in response to the stimulus of water is called
  - A. hydrotropism
  - B. geotropism
  - C. Phototropism
  - D. thigmotropism
9. The overall reaction in glycolysis can be summarised as
  - A.  $C_6H_{12}O_6 \rightarrow 2C_3H_4O_3 + 4H^+ + 2ATP$
  - B.  $C_6H_{12}O_6 \rightarrow 2C_3H_4O_3 + 4H^+ + 2ATP$
  - C.  $C_6H_{12}O_6 \rightarrow 2C_3H_4O_3 + 4H^+ + 2ADP$
  - D.  $C_6H_{12}O_6 \rightarrow 2C_3H_4O_3 + 4H^+ + 2ADP$
10. The longest bone in the body is the
  - A. humerus
  - B. femur
  - C. scapula
  - D. tibia
11. Which of the following structures is not a skeletal material?
  - A. Chitin
  - B. Cartilage
  - C. Bone
  - D. Muscle
12. The reason why the flow of blood through the capillaries is very slow is
  - A. transportation
  - B. water culture
  - C. diffusion
  - D. Osmosis

- A. because the walls of capillaries are very thin
- B. to avoid high — blood pressure
- C. to ensure that the individual does not get dizzy
- D. to allow adequate time for exchange of materials

13. Which of the following groups of organisms has kidney as their excretory organ?
- A. Fishes, amphibians, birds, man
  - B. Fishes, amphibians, annelids, insects
  - C. Fishes, reptiles, birds, tapeworms
  - D. Fishes protozoans, amphibians, man
14. Which of the following features is not a characteristic of arteries? Arteries
- A. possess valves at intervals throughout their length.
  - B. have thick muscular and elastic walls
  - C. carry blood away from the heart
  - D. transport oxygenated blood with the exception of the pulmonary artery.

**The graph below shows the results of a laboratory investigation which measured the body temperatures of a lizard and a bird under changing artificial conditions. Use to answer questions 15 and 16.**



15. Which of the statements below is valid?
- A. The bird's blood was always warmer than that of the lizard.
  - B. The body temperature of the bird varied less than that of the lizard during changes in environmental temperature.

- C. The body temperature of the bird remained constant despite changes in environmental temperature.
- D. The body temperature of the lizard was always close to that of the environmental temperature.

16. What physiological term can be used to describe the regulation of the body temperature of the lizard?
- A. Homeostasis
  - B. Homeothermy
  - C. Poikilothermy
  - D. Osmoregulation
17. The reason why hospitals use saline solutions as drip instead of water is
- A. because salt is a preservative
  - B. to prevent contamination of the body
  - C. to maintain the composition of body fluids
  - D. to increase the number of blood cells
18. The part of the ear which contains nerve cells sensitive to sound vibrations is the
- A. cochlea
  - B. ampulla
  - C. tympanum
  - D. malleus
19. Spectacles with convex lenses correct long-sightedness by
- A. converging the Light rays before they enter the eye
  - B. diverging the light rays before they enter the eye
  - C. reducing light intensity before it enters the eye
  - D. increasing light intensity before it enters the eye
20. A seed of a flowering plant can best be described as
- A. radicle and plumule
  - B. the developed ovule
  - C. the embryo and endosperm
  - D. developed ovary
21. Which of the following processes removes carbon from the atmosphere?
- A. Putrefaction

- B. Photosynthesis
- C. volcanic eruption
- D. Burning fuels

22. Which of the following cycles involves the process of precipitation and transpiration?
- A. Water cycles
  - B. Carbon cycle
  - C. Nitrogen cycle
  - D. oxygen cycle
23. What is the critical limiting factor for plants below the photic zone in an aquatic ecosystem?
- A. Availability of nutrients
  - B. Availability of water
  - C. intensity of light
  - D. Carbon dioxide concentration
24. Which of the following instruments is used to estimate the number of plants in a habitat?
- A. Pooter
  - B. Pitfall trap
  - C. Quadrat
  - D. Sweep net
25. Which of the following statements is true about sandy soil? It
- A. has limited air space
  - B. is light and easy to dig
  - C. drains slowly
  - D. is heavy and poorly aerated
26. Which of the following organisms is a primary consumer?
- A. Dog
  - B. Sheep
  - C. Grass
  - D. Fungus

**Study the diagram of a food chain shown below and use it to answer question 27 and 28.**

P→Q→R→S→T

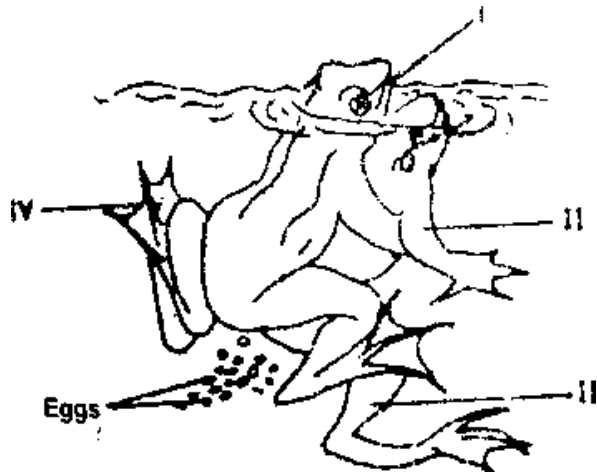
27. The organism designated P in the food chain above is normally sustained by energy from
- A. sunlight
  - B. carbohydrates

- C. green plants
- D. mineral salts

28. Which of the following statements best describes the organism designated R? It
- A. feeds on S.
  - B. is a primary consumer.
  - C. is a producer as well as a consumer
  - D. is a secondary consumer
29. Which of the following diseases is not hereditary?
- A. Albinism
  - B. Scabies
  - C. Haemophilia
  - D. Colour blindness
30. The immediate product of meiosis in flowering plants is the
- A. sporophyte
  - B. gametophyte
  - C. zygote
  - D. pollen grains
31. DNA in eukaryotic cells is contained in the
- A. central vacuole
  - B. nucleus
  - C. lysosome
  - D. golgi body
32. A man who is heterozygous for the disease haemophilia marries a woman who is double recessive for haemophilia. What percentage of their offspring would have the disease?
- A. 0%
  - B. 25%
  - C. 50%
  - D. 75%
33. Cytokinesis of mitosis is a process that ensures that
- A. each daughter cell gets the necessary organelle
  - B. there is distribution of a complete set of genes into each daughter cell.
  - C. daughter cells inherit new genetic combinations.
  - D. worn out organelles are excluded from daughter cells

34. An animal which is active during the day is known as a
- nocturnal animal
  - diurnal animal
  - terrestrial animal
  - homoatomic animal
35. Evidence of evolution include the following except
- fossil records
  - comparative anatomy
  - mutation of genes
  - geographical distribution of organisms

**Use the diagram below to answer question 36 and 37.**



36. The diagram shows that the organisms are
- hermaphrodite
  - viviparous
  - oviparous
  - ovoviparous
37. The breeding posture illustrated in the diagram is known as
- mating
  - amplexus
  - courtship display
  - reproductive swimming
38. An accurate identification of a rapist can be carried out by it conducting a
- RNA analysis
  - DNA analysis
  - blood group test

D. behavioural traits test

39. A boy who is fond of swimming in a pond finds himself passing urine with traces of blood. He is likely to have contracted
- schistosomiasis
  - onchocerciasis
  - poliomyelitis
  - salmonellosis
40. The flippers of a whale and the fins of a fish are examples of
- divergent evolution
  - coevolution
  - continuous variation
  - convergent evolution

1. 2  
6.B  
27.A  
28.D  
29.B  
30.D  
31.B  
32.C  
33.B

## ANSWER KEYS

1. D
2. A
3. B
4. A
5. D
6. B
7. C
8. A
9. B
- 10.B
- 11.D
- 12.D
- 13.A
- 14.A
- 15.C
- 16.C
- 17.A
- 18.A
- 19.A
- 20.B
- 21.B
- 22.A
- 23.C
- 24.C
- 25.B
- 26.
- 27.A
- 28.D
- 29.B
- 30.D
- 31.B
- 32.C
- 33.B
- 34.B
- 35..D
- 36.C
- 37.A
- 38.B
- 39.A
- 40.A